List of Selected Papers

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The following are the key papers that the report *Tabular Data Understanding* draws upon and that form the basis for discussion, organized by the two pillars around which the report is structured.

Foundations and Representations

[S1] X. Wu, A. Ritter, and W. Xu, **"Tabular Data Understanding with LLMs: A Survey of Recent Advances and Challenges"**, *arXiv:2508.00217*, 2025.

[S2] Y. Sui, M. Zhou, M. Zhou, S. Han, and D. Zhang, **"Table Meets LLM: Can Large Language Models Understand Structured Table Data? A Benchmark and Empirical Study"**, in *Proceedings of the 17th ACM International Conference on Web Search and Data Mining (WSDM)*, 2024.

Schema Understanding (Text-to-SQL)

[S3] T. Yu, R. Zhang, K. Yang, M. Yasunaga, D. Wang, et al., **"Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task"**, in *Proceedings of EMNLP*, 2018.

[S4] J. Li, B. Hui, G. Qu, J. Yang, B. Li, et al., **"Can LLM Already Serve as a Database Interface? A Big Bench for Large-Scale Database Grounded Text-to-SQLs (BIRD)"**, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[S5] M. Pourreza, H. Li, R. Sun, Y. Chung, S. Talaei, et al., **"CHASE-SQL: Multi-Path Reasoning and Preference Optimized Candidate Selection in Text-to-SQL"**, *arXiv:2410.01943*, 2024.

[S6] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, **"Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection"**, *arXiv:2310.11511*, 2023.

Data Understanding

[S7] P. Pasupat and P. Liang, **"Compositional Semantic Parsing on Semi-Structured Tables"**, in *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2015.

[S8] Z. Cheng, H. Dong, Z. Wang, R. Jia, J. Guo, et al., **"HiTab: A Hierarchical Table Dataset for Question Answering and Natural Language Generation"**, in *Proceedings of ACL*, 2022.

[S9] W. Chen, H. Wang, J. Chen, Y. Zhang, H. Wang, et al., **"TabFact: A Large-Scale Dataset for Table-Based Fact Verification"**, in *International Conference on Learning Representations (ICLR)*, 2020.

Graph Representation of Tabular Data

[S10] Q. Li, C. Huang, S. Li, Y. Xiang, D. Xiong, and W. Lei, **"GraphOTTER: Evolving LLM-Based Graph Reasoning for Complex Table Question Answering"**, in *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*, 2025.

[S11] X. He, Y. Tian, Y. Sun, N. V. Chawla, T. Laurent, Y. LeCun, X. Bresson, and B. Hooi, **"G-Retriever: Retrieval-Augmented Generation for Textual Graph Understanding and Question Answering"**, in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[S12] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, et al., **"From Local to Global: A Graph RAG Approach to Query-Focused Summarization"**, *arXiv:2404.16130*, 2024.